Xiangrui Liu

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Education

Arizona State University — Tempe, USA
Ph.D. of Computer Science — *2024 - now*
Supervisor: *Dr. Yezhou Yang*

University of British Columbia — Kelowna, Canada
Master of Applied Science in Electrical Engineering — *2022 - 2024*
Supervisor: *Dr. Julian Cheng & Dr. Shan Du*

University of British Columbia — Kelowna, Canada
Bachelor of Applied Science — *2016 - 2020*

Research Experience

AI Generated Video Forensics

Microsoft, USA — *Jan 2026–now*

- Conduct research in video forensics, focusing on detecting AI generated and manipulated video content.
- Curated a large scale benchmark for AI generated video (AIGV) detection spanning SOTA generative and editing models, evaluating detectors under real world distribution shifts.
- Developed a Mixture of Experts detector that fuses three orthogonal forensic signals (pixel level frequency artifacts, feature space temporal dynamics, and spatial forensic attention), achieving new SOTA performance.
- Developing a reasoning VLM that detects AIGV by localizing physical and logical anomalies across a multicategory taxonomy, producing interpretable explanations.
- Training the VLM with SFT on chain of thought data, followed by reinforcement learning with evolving rubrics as rewards to improve reasoning quality and robustness.

Identity Preserved Long Video Editing

Pixocial Technology, USA — *Oct 2025–now*

- Developing a diffusion-based video editing framework focused on maintaining temporal consistency and identity preservation across long sequences.
- Analyzing facial feature degradation over time in latent space.
- Designing a metric that measures identity preservation.
- Applying feature anchoring and researching the use of self-forcing to improve identity preservation.
- Designing a LoRA adapter to align the latent distribution of edited characters with the original subject's manifold.

Virtual Insomnia Patient Language Model

Arizona State University, University of Arizona, University of North Texas, USA — *Jan 2026–May 2026*

- Develop an AI-based virtual patient to support therapist training.
- Architecting a real-time RAG pipeline to ensure medical accuracy and consistency across complex, high-stakes insomnia diagnostic scenarios.
- The AI patient will be capable of real-time Q&A across a wide range of insomnia-related scenarios.

Video Relighting Model

Pixocial Technology, USA — *May 2025–Oct 2025*

- Develop a novel training-free framework for high-resolution video relighting model that can be adapted to different video diffusion backbones.
- The framework integrates a lightness-prior–anchored diffusion scheme, a Hybrid Motion-Adaptive Lighting Smoothing Filter, and a LAB-based Detail Fusion module to achieve flicker-free and detail-preserving results.
- The framework achieved 80% improvement in lighting stability and 56% improvement in detail preservation compared to prior methods, setting the new SOTA.

Hallucination in Vision-language Model

Intel Lab, USA — *Nov 2024–April 2025*

- Investigate the hallucination problem in VLM from a cognitive bias perspective.
- Develop and curate a benchmark to systematically evaluate model reliability, and conduct extensive experiments on SOTA models.
- Analyze underlying causes of hallucination and provide insights toward mitigating the issue.

Marine Mammal Sound Classification

UBC, Canada — *July 2022–Sept 2023*

- Developed a high-speed marine mammals' sound classifier that obtains state-of-the-art accuracy and efficiency.
- Increased the robustness using speaker embedding. Propose a temperature cooling knowledge distillation strategy to train a pruned student model.

Publications[1]

[**WACV 2027**]: **X. Liu**, et al. "How Scaling VLMs Trades Cognitive Biases." (arxiv.org/abs/2507.03123)

[**CVPR 2026**]: **X. Liu**, H. Li, Y. Yang. "Hi-Light: A Path to High-fidelity, High-resolution Video Relighting with a Novel Evaluation Paradigm.

[**EMNLP 2026**]: S. Hegde, **X. Liu**, et al. "Where Hallucinations Live: A Cross-Architecture Circuit in VQ-Tokenized Vision-Language Models."

[**NeurIPS 2025, UrbanAI**]: **X. Liu***, L. Chao*, M. Shivakoti, T. Kutralingam, Y. Yang, H. Wei. "ShadeBench: A Benchmark Dataset and Method for Shade Simulation in Sustainable Society."

[**IJCAI 2025**]: **X. Liu***, L. Chao*, M. Shivakoti, Y. Yang, H. Wei. "DeepShade: Enable Shade Simulation by Text-conditioned Image Generation."

[**AAAI 2024**]: **X. Liu**, X. Liu, S. Du, J. Cheng. "Hear You Say You: An Efficient Framework For Marine Mammal Sounds' Classification."

[**Neurocomputing**]: A. Salari, A. Djavadifar, **X. Liu**, and H. Najjaran, "Object recognition datasets and challenges: A Review."

[**NeurIPS 2021**]: J. Bin, R. Zhang, C. Zhang, **X. Liu**, S. Du, E. Blasch, and Z. Liu, "Improved Object Detection in Thermal Imaging Through Context Enhancement and Information Fusion: A Case Study in Autonomous Driving."

[**In Submission**]: **X. Liu**, et al. "Foundation Model in Biomedicine." (arxiv.org/abs/2503.02104)

[**In Submission**]: **X. Liu**, et al. "ClinRevise: Chest X-Ray Report Refinement from Denoising Feedback."

[**In Submission**]: **X. Liu**, et al. "Evaluating AI-Manipulated Video Detectors under Real-World Distribution Shifts"

[**In Submission**]: C. Huang*, **X. Liu***, et al. "Spatial World Model: Spatially Structured Latent Dynamics and Planning"

Professional Experience

Graduate Research Associate

Arizona State University, USA *Aug 2024–Now*

- Researched on video generation and video editing.
- Researched on AI safety and robustness in computer vision AI systems.
- Researched deterministic action in world models.

AI Research Intern

GE Precision Healthcare LLC, USA *May 2026–Aug 2026*

- Proposed a novel cycle reward tuning paradigm for medical image report generation, leveraging image reconstruction quality as a reinforcement learning reward.
- Trained a diffusion-based medical image generator and integrated it with a vision language model to create a closed loop report & image optimization framework.
- Developed a cycle consistency reward that incentivizes generation of detailed reports capable of faithfully reconstructing corresponding medical images.
- Applied reinforcement learning to optimize report generation using rewards derived from diffusion model reconstructions, improving image report semantic alignment.

Research Assistant

UBC, Canada *May 2023–May 2024*

- Conducted research on 3D face generation, students' engagement analysis in the classroom and AIGC detection.
- Designed a deep learning course for graduate students: PowerPoint slides, assignment questions, exam papers, and projects.

Research Intern

AISpeech Co. Ltd., China *Aug 2021–Nov 2021*

- Developed a voice activity detection model for multiple scenarios.
- Introduced speaker adaptation to the model. Improved the 1600-hours-audio model's accuracy by 3% absolute value.

Research Intern

Suzhou Privacy Technology Co. Ltd., China *Sept 2020–April 2021*

- Researched and developed a multi-lingual multi-class speech emotion classifier.
- The feature engineering solves the imbalanced problem and reduces the model parameters by 85%.
- Compared to the Baseline of MEC2017, the classifier has 36% enhanced in precision and 123% in recall.

Academic Service

- ECCV reviewer - 2026
- WWW reviewer - 2026
- WACV reviewer - 2026
- CVPR reviewer - 2025
- ICCV reviewer - 2025
- NeurIPS reviewer - 2025

[1] Full publication list can be found in my Google Scholar.